SCOREBOARD – Hadamard 14052 Paley-I (2026-10-08)

Cutoff: 2026-10-08T07:29:16Z (~3:29AM ET)

Object	Status	max|HHˆT-nI|	Wall (s)	SPK
H_14052 Paley-I (q=14051)	PASS	0	2402.3848	**IN SPK** (Paley-I)
H_14052 Paley-II (q=7025)	FAIL/SKIP	–	–	q=7025=5ˆ2*281 NOT prime; ===1 mod 4 but composite
H_14052 Kronecker (4*3513)	FAIL/SKIP	–	–	3513===1 mod 4 NOT Hadamard-admissible; 3513=3*1171 (odd composite)

Units: 640/640 * row blocks 110/110 * overall_pass=True * wall_total=2440.8133s
Gap 2052: **OPEN** * clay_closed=false * GATE-04 **LOCKED** * conjecture **OPEN**
IN SPK via Paley-I; Miyamoto digression skipped; exclude Miyamoto-only 8948/8644/8516/...; prior SPK 10224+10212+10164+10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9884 retained

Next optional: **q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3ˆ271*317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59ˆ2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2ˆ2*59ˆ2 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2ˆ2*3*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3ˆ2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2ˆ2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3^2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2ˆ2*3^2*389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13ˆ2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2ˆ2*3*7167 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14052: q2=7057->n=14116 (q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2ˆ2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII); Paley-I DIRECT 14072/14084/14088/14108 come first) | Kronecker-only (no direct Paley route; note-only, NOT on box): none in 14052..14072 (14064=4*3516, 3516=2ˆ2*3*293**

===0 mod 4 order-admissible but H_3516 has no direct classical Paley route: 3515=5*19*37, 1757=7*251 composite); beyond: 14112=4*3528 (3528=$2^{3^{3^{\hat{}}2}}$7^{2}2 ===0 mod 4; H_3528 via Paley-I q=3527 prime ===3 mod 4) | next Paley-I DIRECT: q=14071->n=14072 (prime ===3 mod 4; 14057 prime ===1 mod 4 skipped; no prime ===3 mod 4 in 14052..14070; FIRST n===0 mod 4 after 14052 with any direct classical Paley route (14056/14060/14064/14068 none); Paley-II q=7035=3*5767 NOT prime (===3 mod 4 and composite) -> FAIL/SKIP; Kronecker 14072=4*3518; 3518=2*1759 ===2 mod 4 -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3518); 14072=2^3*1759 ===8 mod 16 = 13880/13832/13752-like shape (q2 composite ===3, m===2 mod 4)) (or gap 2052 / Track B)